Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech (IT) Examination

March 2020

IT341/IT302.02/IT302.01/IT302 Design & Analysis of Algorithms

Date: 14.03.2020, Saturday **Time: 10.00 a.m. To 01.00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 **Attempt the following questions.** **[07]**

1. Define Time complexity of an algorithm. Explain best case and worst case analysis of an algorithm. [03]
2. Derive Big Oh time complexity for following code. [02]

```
k=2;
while (k<=m)
{
    k = k^3;
}
```

3. What is Divide and Conquer approach? Write general recurrence relation for divide and conquer approach. [02]

Q-2 (a) Discuss the importance of asymptotic notations in algorithm analysis. Explain different asymptotic notations. **[06]**

Q-2 (b) Compare Greedy approach and Dynamic Programming approach. **[04]**

Q-2 (c) Which design approach is used for Binary Search algorithm? Explain and analyze best case, average case and worst case time complexity of Binary Search algorithm. **[04]**

OR

Q-2 (b) Find the optimal solution for the fractional knapsack problem making use of greedy approach. Consider below data. **[04]**

Item	I1	I2	I3	I4	I5
Weight	5	10	15	22	25
Value	30	40	45	77	90

Q-2 (c) Apply Quicksort to sort following data in ascending order. Select first element as a pivot element. **[04]**

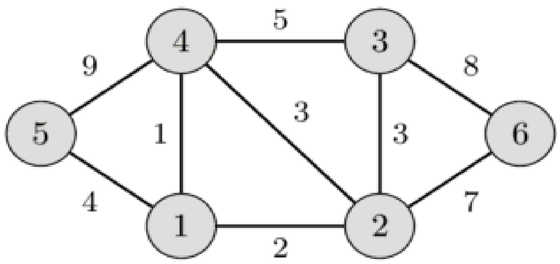
4, 20, 10, 80, 7, 30

Q-3 **Attempt the following questions. (Any Two)** **[14]**

1. What is N Queen Problem? Draw a pruned state space tree for 4-Queen's problem up to first two solution using Backtracking approach.
2. Derive mathematical equation to find time complexity of following algorithm for Best case and Worst case.

```
Function(A: An Array)
for(i=0; i<n; i++)
{   exchange = 0;for(j=0; j< n-i; j++)
      {
            if(A[j] > A[j+1])
                  {
                    Swap(A[j],A[j+1]);
                    exchange = exchange +1;
                  }
      }
   if(exchange ==0)
            break;
}
```

3. What is Minimum Spanning Tree? Find minimum spanning tree for a given graph using Kruskal's algorithm.



SECTION-II

Q-4 **Attempt the following questions.** **[07]**

1. True/False. [02]
   Dynamic Programming is Bottom-up technique. Justify your answer.
2. Arrange the given terms in increasing order of their time complexity. [02]
   O($\sqrt{n}$), O($n^{(3/2)}$), O($lg(lgn)$) , O(n), O($n^2 lgn$), O(2^n), O($n!$), O($nlgn$)
3. List out similarities and differences between Branch & Bound and Backtracking Approach. [03]

Q-5 (a) State Master Theorem and Solve following recurrence using Master Theorem. **[05]**

$$T(n) = 4T(n/2) + n^2$$

Q-5 (b) Explain Naïve String Matching algorithm. **[05]**

Q-5 (c) Explain P and NP-Hard problem. **[04]**

OR

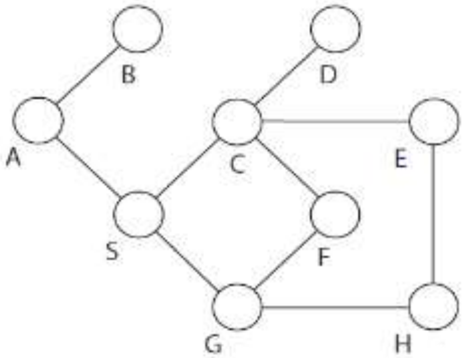
Q-5 (b) Solve using Recurrence Tree Method. **[05]**

$$T(n) = 4T(n/2) + n$$

Q-5 (c) Explain Rabin-Karp string matching algorithm **[04]**

Q-6 **Attempt the following questions. (Any Two)** **[14]**

1. Define Graph. Which are the different graph traversal techniques. Compare them and apply them on the given graph with source vertex S.



2. Using Dynamic programming, find an optimum order of a matrix chain product whose sequence of dimensions is <5, 10, 3, 5, 6, 2>.

3. Given two sequences of characters: P=<XMJYAUZ> and Q=<MZJAWXU>. Obtain the longest common subsequence.

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